Two Exact Values and a New Upper Bound for Binary 2-Cover-Free Families

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Abstract

Let $a(n)$ be the largest size of a family of subsets of an n-set in which no member is contained in the union of two other distinct members; equivalently, $a(n)$ is the maximum number of columns in an n-row binary 2-disjunct matrix. This is OEIS A286874. We prove

$$a(15) = 42, \qquad 48 \leq a(16) \leq 54, \qquad a(17) = 68.$$

The common upper-bound tool is a mixed-weight private-subset inequality on random maximal chains. At $n = 17$ it matches the 68 blocks of an $S(3, 5, 17)$. At $n = 15$ it is combined with hereditary row cells, the complete classification of the 28-member extremal families on 14 points, and finite profile and compatibility-graph certificates. At $n = 16$, an exact enumeration of 97 chain-feasible profiles is eliminated by sharing the chain slack between a weight-5 packing obstruction and a weight-7 triple-union obstruction. A standard-library-only reproducibility archive contains all constructions, input data, scripts and machine-readable certificates.

1 Introduction

A family $\mathcal{F} \subseteq 2^{[n]}$ is *2-cover-free* if for every three distinct $A, B, C \in \mathcal{F}$,

$$C \not\subseteq A \cup B.$$

In matrix language, for every target column C and distinct source columns A, B, some row has pattern $(C, A, B) = (1, 0, 0)$. Such matrices are also called binary 2-disjunct matrices or superimposed codes; see Kautz and Singleton [5] and Erdős, Frankl and Füredi [2]. The sequence of maxima is OEIS A286874 [6].

The historical 28-vector lower-bound construction at $n = 14$ was recorded by Dmitry Kamenetsky [4]; Steinar H. Gunderson later proved exactness and supplied the complete classification [1]. The previously recorded exact values therefore ended at $a(14) = 28$, while Gunderson's Gunderson's OEIS attachment gave lower bounds $42, 48, 52$ at $n = 15, 16, 17$ [3]. The complete classification of the $a(14)$ extremals, also contributed by Gunderson as an attachment to OEIS A303977 [1], is used below as a finite input. Our principal results are the following.

Theorem 1. *The unrestricted binary 2-cover-free maxima satisfy*

$$a(15) = 42, \qquad 48 \leq a(16) \leq 54, \qquad a(17) = 68.$$

The $S(3, 5, 17)$ lower construction is classical: it is the inversive plane of order four. Inversive planes $S(3, q + 1, q^2 + 1)$ exist for prime-power q; see Phelps [7]. Zhu gave an explicit pair of orthogonal $S(3, 5, 17)$ systems [8]. We found the particular 68-block list distributed here with a CP-SAT packing search seeded with Gunderson's 52-block witness [3]; it is checked independently at the incidence level, so neither the solver nor the seed is used in the proof of validity.

2 The private-chain inequality

For $F \in \mathcal{F}$, call $S \subseteq F$ *private to* F if no other member of $\mathcal{F}$ contains S.

Lemma 2. *For every $S \subseteq F$, at least one of S and $F \setminus S$ is private to F. Private subsets owned by different members are cross-incomparable.*

Proof. If neither side were private, some $G, H \in \mathcal{F} \setminus \{F\}$ would satisfy $S \subseteq G$ and $F \setminus S \subseteq H$, whence $F \subseteq G \cup H$. The two witnesses cannot coincide because a 2-cover-free family with at least three members is an antichain. If P is private to F, Q is private to $G \neq F$, and $P \subseteq Q$, then $P \subseteq G$, a contradiction. $\square$

Choose a uniformly random maximal chain of $2^{[n]}$, equivalently a random permutation of $[n]$. If $|F| = w$, let R_F be the elements of F occurring before the first point outside F. For fixed $S \subset F$, $|S| = s < w$,

$$\Pr(R_F = S) = \mu_{n,w}(s) = \frac{n-w}{n-s}\binom{n}{s}^{-1},$$

and $\Pr(R_F = F) = \binom{n}{w}^{-1}$. The private subsets of F form an upset, so the chain meets one exactly when R_F is private.

Pair every subset of F with its complement in F. At least one side is private by Lemma 2, and the larger side has the smaller μ-mass. Define

$$\rho(n, 2t-1) = \sum_{s=t}^{2t-1} \binom{2t-1}{s} \mu_{n,2t-1}(s),$$
$$\rho(n, 2t) = \frac{1}{2}\binom{2t}{t}\mu_{n,2t}(t) + \sum_{s=t+1}^{2t} \binom{2t}{s}\mu_{n,2t}(s).$$

Theorem 3 (Mixed-weight private-chain bound)**.** *Every 2-cover-free family with at least three members satisfies*

$$\sum_{F \in \mathcal{F}} \rho(n, |F|) \leq 1.$$

Proof. For each F, the preceding complementary-pair argument lower-bounds the probability that the random chain meets a private subset owned by F by $\rho(n, |F|)$. By Lemma 2, a totally ordered chain cannot meet private subsets owned by two different members. The events are therefore disjoint. $\square$

All values of μ and ρ used below are evaluated as reduced rational numbers by `scripts/common.py`.

3 The exact value at 17

At $n = 17$, direct evaluation gives

$$\min_{1 \leq w \leq 17} \rho(17, w) = \frac{1}{68},$$

with equality only at $w = 5, 7$. Theorem 3 therefore gives $a(17) \leq 68$ for arbitrary mixed weights.

The file `constructions/a17_68.txt` contains 68 weight-5 blocks found as described above. The replay verifies that distinct blocks meet in at most two points and that their $68\binom{5}{3} = 680 = \binom{17}{3}$ triples are all distinct. Hence they form an $S(3, 5, 17)$ and are 2-cover-free: a third 5-block has at most four points in the union of any two others. Thus $a(17) = 68$.

4 The exact value at 15

Gunderson's construction in `constructions/a15_42.txt` [3] proves $a(15) \geq 42$. We exclude 44 and then 43 members.

4.1 An incidence obstruction at 44

The following fifteen rational inequalities are checked directly:

$$w \leq 546\rho(15, w) - 7 \qquad (1 \leq w \leq 15),$$

with equality only at $w = 5, 7$. If $m = |\mathcal{F}|$ and $I = \sum_{F \in \mathcal{F}} |F|$, Theorem 3 gives

$$I \leq 546 - 7m. \tag{1}$$

The members zero on any fixed row, after deleting that row, form a 2-cover-free family on 14 rows and hence number at most $a(14) = 28$. Every row has degree at least $m - 28$, so

$$I \geq 15(m - 28). \tag{2}$$

At $m = 44$, these give $I \leq 238$ and $I \geq 240$. Therefore $a(15) \leq 43$.

4.2 The minimum-weight-3 branch

Assume a 43-member family has a minimum member C of size three. The nonempty zero traces $C \setminus B$, over the other 42 members B, are pairwise intersecting. No coordinate of C occurs in more than 28 traces. A singleton trace would force all 42 traces through one point, so every trace has size at least two. The trace-incidence total is consequently at least 84 and at most $3 \cdot 28 = 84$. Equality forces fourteen copies of each 2-subset of C.

The union of any two resulting 14-member trace classes has a constant-zero support row. Deleting it produces an extremal 28-member family on 14 rows with two complementary coordinate rows. Gunderson's complete A303977 input [1] contains 788 extremal classes. The replay verifies every class against the cover-free definition and finds no complementary row pair. Thus minimum weight three is impossible.

4.3 The degree-15 row and the seven profiles

Now all members have weight at least four. Equation (2) gives row degree at least 15. If a row had degree 15, its 28-member zero-cell would be an extremal family on 14 rows. The verified classification consists of 787 all-weight-3 families and one all-weight-5 family, so the zero-cell must be the unique weight-5 extremal.

Fix that extremal and exhaust the 2^{14} residual words for each of the 15 one-cell members. After all conditions involving the fixed family and at most two candidate members are imposed, 98 candidates remain. Their exact compatibility graph has clique number 14. This graph omits constraints among three candidate members, so it is a relaxation; the required 15-clique cannot exist. Every row therefore has degree at least 16, and $I \geq 240$.

Exact integer enumeration of the global chain budget, minimum weight four, 43 members and $I \geq 240$ leaves precisely the seven profiles

$$5^{28}7^{15},\ 5^{29}7^{14},\ 5^{29}6^{1}7^{13},\ 5^{29}6^{2}7^{12},\ 5^{30}7^{13},\ 5^{30}6^{1}7^{12},\ 5^{31}7^{11}8^{1}.$$

For a weight-5 member, a nonprivate triple costs

$$\delta_5 = \mu_{15,5}(2) - \mu_{15,5}(3) = \frac{1}{182}$$

above its minimum chain mass. For a weight-7 member, a nonprivate 4-subset costs

$$\delta_7 = \mu_{15,7}(3) - \mu_{15,7}(4) = \frac{2}{2145}.$$

If the profile slack permits at most q_5, q_7 such incidences, then at least $10b_5 - q_5$ triples are private to the b_5 weight-5 members and occur in no weight-7 member.

Let r_Q be the number of weight-7 members through a 4-set Q. Incidences with $r_Q \geq 2$ are nonprivate, giving $\sum_Q \binom{r_Q}{2} \leq \binom{q_7}{2}$. Since $\binom{r}{3} \leq 1 + 3\binom{r}{4}$ for $0 \leq r \leq 7$, two-term inclusion–exclusion lower-bounds the union of the triples in the weight-7 members by

$$35b_7 - \binom{b_7}{2} - 3\binom{q_7}{2}. \tag{3}$$

For the seven profiles, these lower bounds are

$$420, 390, 377, 354, 314, 324, 330,$$

while the numbers of triples available outside the private weight-5 triples are only

$$175, 165, 165, 165, 156, 156, 145.$$

All cases are impossible, proving $a(15) = 42$.

5 The new upper bound at 16

Gunderson's file `constructions/a16_48.txt` [3] gives $a(16) \geq 48$. The unique minimum chain mass at $n = 16$ is $\rho(16, 5) = 1/56$. Equality with 56 members would force all members to be 5-sets with all triples private, hence an $S(3, 5, 16)$. Its point replication number would be

$$\frac{\binom{15}{2}}{\binom{4}{2}} = \frac{35}{2},$$

which is impossible. Thus $a(16) \leq 55$.

At 55 members, exact enumeration of the chain budget leaves 97 integer weight profiles, supported on weights 3 through 9. A 5-uniform family whose triples are private is a 3-$(16, 5, \leq 1)$ packing. Through each point there are at most

$$\left\lfloor \frac{15}{4} \left\lfloor \frac{14}{3} \right\rfloor \right\rfloor = 15$$

blocks, so it contains at most $16 \cdot 15/5 = 48$ blocks. If q_5 weight-5 triple incidences are nonprivate, deleting at most q_5 affected blocks gives the necessary condition $b_5 - q_5 \leq 48$. Here

$$\delta_5 = \mu_{16,5}(2) - \mu_{16,5}(3) = \frac{11}{2184}.$$

For weight-7 members, equation (3) applies with

$$\delta_7 = \mu_{16,7}(3) - \mu_{16,7}(4) = \frac{3}{3640}.$$

The weight-7 triple union must fit among $\binom{16}{3} - (10b_5 - q_5)$ triples. Crucially, both defect counts spend the same profile slack:

$$q_5 \frac{11}{2184} + q_7 \frac{3}{3640} \leq \text{slack}.$$

Every integer pair (q_5, q_7) for every one of the 97 profiles is checked in `scripts/certify_a16_upper.py`; none meets all three necessary conditions. Therefore $a(16) \leq 54$, proving

$$48 \leq a(16) \leq 54.$$

6 Reproducibility

The archive contains the three construction witnesses, the complete A303977 classification input, standard-library Python certifiers, deterministic JSON outputs, and SHA-256 hashes. Running

```
python3 scripts/replay_all.py
```

from the repository root verifies the following independently:

- every construction against the original OR-containment definition;
- all 788 classified $a(14)$ extremals and the absence of complementary rows;
- the 98-vertex graph and clique number 14;
- the seven final $n = 15$ profiles and their exact rational slacks;
- all 97 $n = 16, m = 55$ profiles and every joint defect allocation;
- all mixed-weight chain masses used at $n = 15, 16, 17$.

No floating-point optimization, external solver or timeout is used in a final claim. The role and provenance of every file are described in `SUPPORTING_FILES.md`.

A Complete ancillary-file inventory

This appendix records every file that supplies a construction, enters an upper-bound reduction, records its output, or allows the result to be replayed. Paths are relative to the repository root.

A.1 Mathematical input and witnesses

`sources/a14_extremals_A303977.txt`. The sole external finite-classification input. It contains all 788 equivalence classes of 28-member extremal families on 14 points from OEIS A303977. The attachment and the $a(14)$ classification were contributed by Steinar H. Gunderson [1]; A303977 itself was created by Zhao Hui Du. Kamenetsky's earlier A286874 attachment contains the historical 28-vector lower-bound construction [4]. The file is used twice in the $n = 15$ upper bound: to exclude the minimum-weight-3 trace branch by checking that no class has complementary rows, and to identify the single weight-5 class used in the degree-15-row branch. The replay parses and directly verifies every class before using either fact. Its provenance and original SHA-256 hash are recorded in `sources/README.md`.

constructions/a15_42.txt. Steinar H. Gunderson's 42-column witness [3], proving $a(15) \geq 42$. It is reproduced verbatim in Appendix B and checked directly against the defining ordered triples of columns.

constructions/a16_48.txt. Steinar H. Gunderson's 48-column witness [3], proving $a(16) \geq 48$. It is reproduced verbatim in Appendix C and checked directly against the definition.

constructions/a17_68.txt. The 68-column witness proving $a(17) \geq 68$. We found this particular list using a CP-SAT search seeded with Gunderson's 52-column witness [3]. Classical existence is due to the inversive-plane construction [7], and an explicit pair was published by Zhu [8]. The list is reproduced verbatim in Appendix D. Besides the direct cover-free check, the replay verifies weight five, pairwise intersection at most two, and exact coverage of all 680 triples, certifying that it is an $S(3, 5, 17)$.

A.2 Executable reductions

scripts/common.py. Shared exact-arithmetic implementation. It defines the reduced rational functions μ and ρ, construction parsers, the direct cover-free predicate, profile generation, and hashing helpers. All certifiers import this file.

scripts/certify_a15_exact.py. Complete executable upper-bound reduction for $n = 15$. It verifies the 42 witness and all 788 classified $n = 14$ families; checks the affine incidence obstruction at 44 members; excludes the minimum-weight-3 branch; reconstructs the 98-vertex degree-15 compatibility relaxation and proves clique number 14; enumerates the seven surviving 43-member profiles; and checks each triple-union contradiction using exact rational slack.

scripts/certify_a16_upper.py. Complete executable proof that a 55-member family on 16 points cannot exist. It checks the 48 witness, the $S(3, 5, 16)$ replication obstruction at 56, enumerates all 97 chain-feasible profiles at 55, and exhausts every shared weight-5/weight-7 defect allocation allowed by the exact chain slack.

scripts/certify_a17_exact.py. Complete executable verification of the $n = 17$ construction and upper bound. It checks the $S(3, 5, 17)$ incidence properties, direct cover-freeness, and $\rho(17, w) \geq 1/68$ for all $1 \leq w \leq 17$.

scripts/replay_all.py. The public replay entry point. It invokes all three certifiers, checks their terminal assertions, and writes the aggregate summary. Running this file is sufficient to reproduce every computational assertion in the paper.

scripts/build_sha256s.py. Integrity utility rather than a mathematical assumption. It deterministically hashes every distributed file except the hash list itself.

A.3 Historical construction-discovery files

These files record how the particular labeled $n = 17$ witness was found. They are not inputs to the lower-bound verification or to any upper-bound proof.

discovery/search_constant_weight.py. The CP-SAT packing-search program used to find the 68-block witness. It selects weight-5 subsets subject to the condition that every triple occurs in at most one selected block. The program requires OR-Tools and parallel search need not reproduce the same labeled solution. Its saved result is therefore verified independently rather than trusted.

discovery/a17_seed_52.txt. Gunderson's 52-vector OEIS witness [3], used as the initial CP-SAT hint. The search first found 55 blocks and then, using that improvement as a new hint, found 68. The overwritten intermediate 55-block file is not used in any final argument.

discovery/README.md. Records the two-stage search history, dependency boundary, a representative command, and the fact that discovery replay is optional. The standard-library certifier, not CP-SAT, establishes the validity and Steiner property of the saved construction.

A.4 Machine-readable certificates and audit documents

certificates/a15_exact.json. Deterministic record of the $n = 15$ replay, including classification counts, graph order and clique number, seven profiles, rational slacks, defect budgets, and final inequalities.

certificates/a16_upper_54.json. Deterministic complete table of the 97 profiles at $n = 16, m = 55$, together with the exact failure of every admissible joint defect allocation.

certificates/a17_exact.json. Deterministic record of the construction incidence checks and all 17 private chain masses used for $n = 17$.

certificates/replay_summary.json. Aggregate pass/fail record written only after all three individual certificates succeed.

SUPPORTING_FILES.md. Short claim-to-file map for reviewers. It adds no mathematical assumption.

docs/PROOF.md. Expanded prose version of the proof and certificate reductions. It adds no input beyond the theorem statements and files enumerated here.

docs/REPRODUCE.md. Exact commands and requirements for an independent replay.

sources/README.md. Provenance and fixed hashes for the A303977 input and three witnesses.

SHA256SUMS. Hash manifest for the complete repository snapshot. It detects accidental changes but is not assumed in any mathematical inference.

The remaining files supply only navigation, citation, licensing and build metadata: `README.md`, `CITATION.cff`, `LICENSE.md`, `Makefile`, `.gitignore`, and the manuscript source, PDF and bibliography. They are not hidden mathematical inputs. Thus the inventory above accounts for every file used to obtain a construction or tighten an upper bound, as well as every derived certificate used to audit those steps.

B Exact 42-member construction at $n = 15$

Each line after the heading is one length-15 binary vector. The list below is included verbatim from `constructions/a15_42.txt`. Construction credit: Steinar H. Gunderson [3].

```
a(15) >= 42:
000000000011111
000000011100011
000000101101100
000001010110100
000001101010001
000001110001010
000010011011000
000100100110010
000110010000110
000110100001001
000111001100000
001000110000101
001010000110001
001010101000010
001011000001100
001100001010100
001100010101000
001101000000011
010001000101001
010010001000101
010010110100000
010011000010010
010100001001010
010100010010001
010101100000100
011000000100110
011000100011000
011001011000000
100001001000110
100010000101010
100010100010100
100011010000001
100100000100101
100100111000000
100101000011000
101000001001001
101000010010010
101001100100000
110000001110000
110000010001100
110000100000011
111110000000000
```

C Exact 48-member construction at $n = 16$

Each line after the heading is one length-16 binary vector. The list below is included verbatim from `constructions/a16_48.txt`. Construction credit: Steinar H. Gunderson [3].

```
a(16) >= 48:
0000000000011111
0000000011100011
0000000101101100
0000001010110100
0000010011011000
0000011100000011
0000100100110001
0000101000101010
0000101111000000
0001000110001001
0001010000110010
0001011000001100
0001100100000110
0001110001000001
0010000110010010
0010010010000101
0010011001100000
0010100001010100
0010110100001000
0011000001001010
0011001000010001
0011100010100000
0100001001001001
0100010001000110
0100010110100000
0100100010001100
0100111000010000
0101000000100101
0101000101010000
0101001010000010
0110000000111000
0110001100000100
0110100000000011
1000001001010010
1000010000101001
1000010100010100
1000101000000101
1000110010000010
1001000011000100
1001001100100000
1001100000011000
1010000000100110
1010000101000001
1010001010001000
1100000010010001
1100000100001010
1100100001100000
1111010000000000
```

D Exact 68-member construction at $n = 17$

Each line after the heading is one length-17 binary vector. The list below is included verbatim from `constructions/a17_68.txt`. These are the 68 blocks of the verified $S(3, 5, 17)$ used in the lower bound. This particular list was found as described in Appendix A; the search was seeded with Gunderson's 52-block witness [3], while Phelps and Zhu supply the classical design references [7, 8].

```
a(17) >= 68:
01000000000001111
00000001100010011
00000010010100011
00101000001000011
10010100000000011
00000100000110101
00000010101000101
00011000010000101
10100001000000101
00100000010011001
10000000001101001
00001100100001001
00010011000001001
01010000001010001
10001010000010001
00110000100100001
01001001000100001
00000101011000001
11000000110000001
01100110000000001
10000000010010110
00010000001100110
00100100100000110
00001011000000110
00011000000011010
00000101000101010
00000000111001010
10100010000001010
01100000000110010
00000110001010010
10001000100100010
11000001001000010
00110001010000010
01001100010000010
01010010100000010
00000001001011100
00101000000101100
00000110010001100
10010000100001100
01001000100010100
00110010000010100
00000001110100100
11000010000100100
01100000011000100
10001100001000100
01010101000000100
```

```
00000010100111000
11000100000011000
01010000010101000
01001010001001000
00110100001001000
10001001010001000
01100001100001000
00001000011110000
10010001000110000
10100000101010000
00010100110010000
01000011010010000
00101101000010000
01000100101100000
00100011001100000
10100100010100000
00011110000100000
10010010011000000
00011001101000000
00101010110000000
10000111100000000
11111000000000000
```

References

[1] Zhao Hui Du. Oeis a303977: Number of inequivalent solutions to the problem discussed in a286874. The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A303977`, 2018. The complete file of all 788 inequivalent solutions for $a(14)$ was contributed by Steinar H. Gunderson in 2026; `https://oeis.org/A303977/a303977.txt`.

[2] Paul Erdős, Peter Frankl, and Zoltán Füredi. Families of finite sets in which no set is covered by the union of two others. *Journal of Combinatorial Theory, Series A*, 33(2):158–166, 1982.

[3] Steinar H. Gunderson. Lower bounds and their solutions for $a(15)$–$a(17)$. Attachment to OEIS A286874, `https://oeis.org/A286874/a286874_3.txt`, 2026. Source of the packaged 42- and 48-vector constructions; its 52-vector construction seeded the search producing the packaged 68-vector system.

[4] Dmitry Kamenetsky. Lower bounds and their corresponding solutions for $a(12)$–$a(16)$. Attachment to OEIS A286874, `https://oeis.org/A286874/a286874.txt`, 2018. Includes the historical 28-vector lower-bound construction at $n = 14$.

[5] William H. Kautz and Roy C. Singleton. Nonrandom binary superimposed codes. *IEEE Transactions on Information Theory*, 10(4):363–377, 1964.

[6] OEIS Foundation Inc. The on-line encyclopedia of integer sequences, a286874. `https://oeis.org/A286874`. Accessed 6 August 2026.

[7] K. T. Phelps. Direct product of derived steiner systems using inversive planes. *Canadian Journal of Mathematics*, 33(6):1365–1369, 1981.

[8] L. Zhu. A pair of orthogonal steiner systems $s(3, 5, 17)$. *Discrete Mathematics*, 31(1):111–113, 1980.